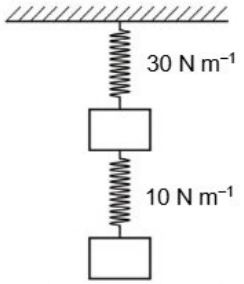
2 Two identical masses are suspended from the ceiling using two massless springs as shown in the diagram below. The upper spring has a spring constant of 30 N m^{-1} and is extended by 2.0 cm from its unstretched length. The lower spring has a spring constant of 10 N m^{-1}.

Both masses are in equilibrium, and both springs obey Hooke's law.



(not drawn to scale)

What is the extension of the lower spring from its unstretched length?

A 1.0 cm

B 1.3 cm

C 3.0 cm

D 6.0 cm